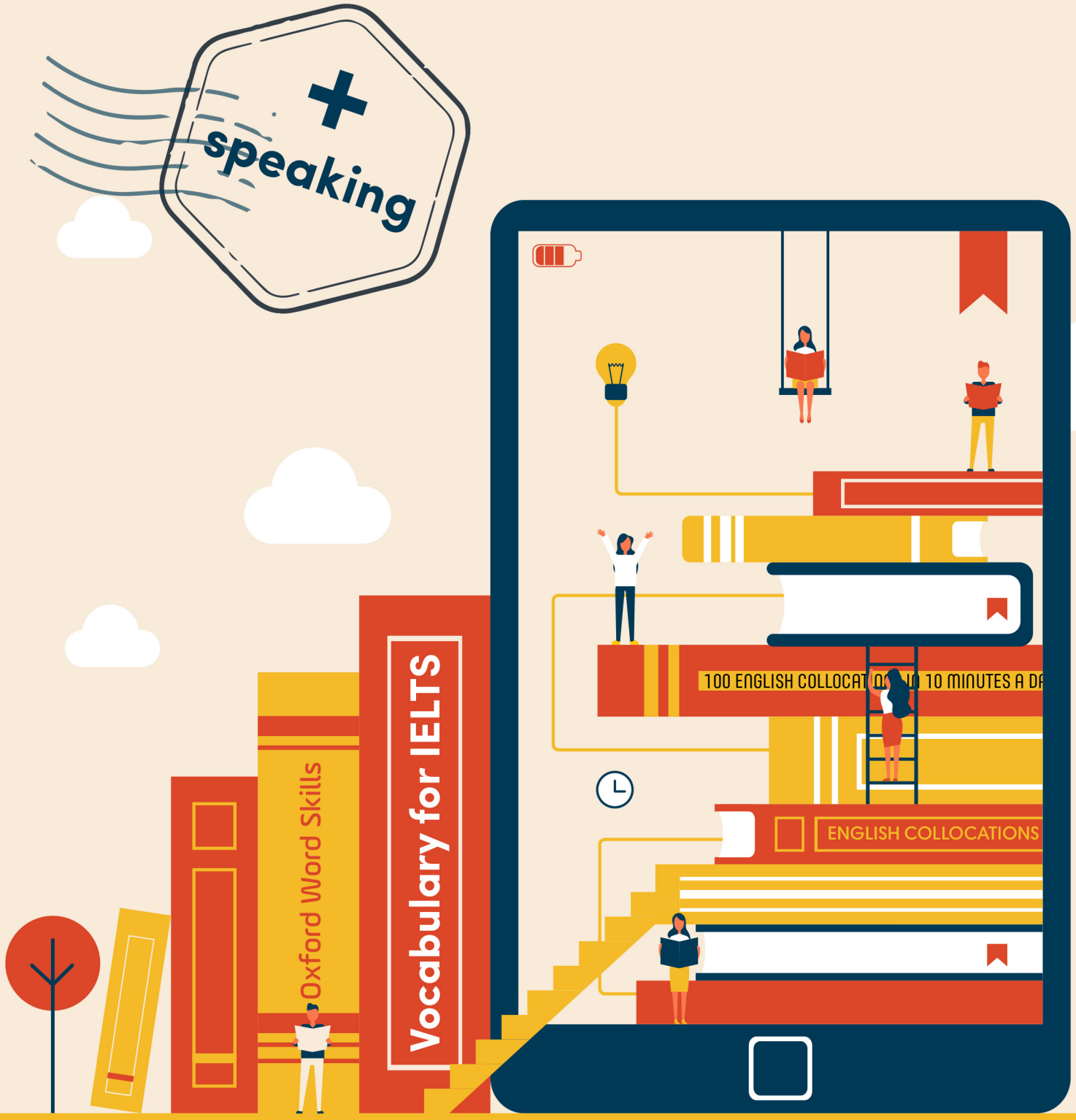
IELTS Up Words!

دوره لغات آیلتس

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Lesson 61: Physics or Physique?

Hello! You are listening to lesson sixty-one of IELTS Up Words! In this lesson, why not get a little geeky and talk about physics? I am not talking about physique, though. That is a different thing.

Physics is the branch of science concerned with the nature and properties of matter and energy. Remember Einstein's theory of relativity? That was a physics equation. What about Newton's second law of motion? Remember what it was? The second law states that the acceleration of an object is dependent upon two variables: the net force acting upon the object and the mass of the object. I know you might argue that without a shadow of a doubt, we don't need to learn about these formulas and physics laws because they are not practical. But I beg to differ, as I think in a few years when humankind starts to inhabit other planets, and as the progress made in science skyrockets, these physics laws will become public knowledge, just like what we know about gravity.

I believe that it won't be long before space tourism becomes a norm and experiencing weightlessness goes right into the top of everyone's bucket list. I mean, it wouldn't be hard to travel through space in a spacecraft or even land on the Moon.

You don't believe me? Just look at the rate scientists are making astronomical discoveries these days. Now, let's get even nerdier and read a passage from "Vocabulary for IELTS Advanced," lesson ten, about our Earth. It is full of words that you might not know. But don't worry; like always, just try to get the gist of it.

You might witness it once, or if you are particularly lucky or very long-lived, perhaps twice. But a total solar eclipse is worth the wait. At the height of totality, the fit of the Sun and the Moon is so perfect that beads of sunlight can only just penetrate the rugged valleys on the lunar surface, creating the stunning diamond ring effect. It is all thanks to a striking coincidence. The Sun is about 400 times as wide as the Moon, but it is also 400 times further away. The two, therefore, look the same size in the sky—a unique

situation among our solar system's eight planets and 188 known moons. Earth is also the only known planet to sustain life. Our Moon is different, but how did it form?

Planetary scientists believe that in the first 100 million years of our solar system, a Mars-sized object smashed into Earth. As Earth spins on its own axis, it has a natural tendency to wobble, owing to the varying pull from other bodies, such as the Sun. The unseen hand of the Moon's gravity gently reduces that wobble, preventing rotational instabilities which would otherwise have caused dramatic changes in Earth's climatic zones over time. Such instabilities would have made it much trickier for life to get started on our planet.

Speaking Tip

The Pronunciation of TH

There are three types of “th” pronunciation:

1) in words like that, these, those, and weather

To pronounce type one, put your tongue between your teeth and try to say “Z.” You should feel a vibration in your throat.

2) in words like thank, thirteen, and thorough

To pronounce type two, put your tongue between your teeth and try to say “T.” You shouldn’t feel any vibration in your throat.

3) in words like Thailand, Thai food, and Thames

The pronunciation of “th” in type three words is exactly like T.

topic-related words and structures	meaning	translation	other structures
geeky	knowledgeable and obsessively interested in something, especially technology	کسی که اطلاعات زیادی دربارهٔ موضوعات علمی و تکنولوژی دارد	nerdy
the nature and properties of matter and energy	the main characteristics of matter and energy	ماهیت و خواص ماده و انرژی	
theory of relativity	A theory of gravity stating that, instead of being an invisible force that attracts objects to one another, gravity is a curving or warping of space. The more massive an object, the more it warps the space around it.	نظریهٔ نسبیت	general relativity; quantum relativity
equation	a statement in math, physics, or chemistry	معادله	equivalence; equality
acceleration	the rate of change in the speed of something over time	شتاب	opposite: deceleration
variables	numbers, amounts, or situations that can change	متغیرها	opposite: constants
the net force	the combination of all the forces that act on an object	نیروی برآیند؛ جمع برداری تمام نیروهایی که روی یک شیء اثر می‌گذارند	total force; resultant force
the mass of the object	a measure of the object's inertial property, or the amount of matter it contains	جرم جسم	weight; size
to skyrocket	to rise or become successful extremely quickly	بالارفتن یا رسیدن به موفقیت در بازه‌ای بسیار کوتاه و سریع	to rocket (informal)

topic-related words and structures	meaning	translation	other structures
public knowledge	the knowledge that is available to anyone	دانش عمومی	general knowledge
space tourism	human space travel for recreational purposes	گردشگری فضایی	space travel
weightlessness	to have no weight; the fact of not being affected by gravity	بی‌وزنی	zero gravity
spacecraft	a vehicle used for travel in space	فضاپیما	spaceship; space shuttle
land on the Moon	come down through the air and alight on the Moon	فرودآمدن روی ماه	Moon landing
astronomical discoveries	really impressive discoveries	اکتشافات عظیم	synonyms: vast; enormous; immense; immeasurable; colossal opposites: imperceptible; minute; minuscule; infinitesimal; microscopic
nerdy	characterized by an obsessive interest in something, especially technology	در توصیف داشتن علاقهٔ بی‌حدومرزی نسبت به چیزی، به‌خصوص تکنولوژی	geeky
solar eclipse	an eclipse in which the Sun is obscured by the Moon	خورشیدگرفتگی	lunar eclipse

topic-related words and structures	meaning	translation	other structures
beads of sunlight	the row of brilliant points of sunlight shining through valleys on the edge of the moon that are seen for a few seconds just before and after the central phase in an eclipse of the sun	پرتوهای نوری که لحظاتی قبل و بعد از خورشیدگرفتگی کامل از کنار لبهٔ قرص تاریک ماه دیده می‌شوند؛ پدیدهٔ حلقهٔ الماس	Baily's beads; the diamond ring effect
penetrate the rugged valleys on the lunar surface	reach the rough, uneven low areas on the Moon's surface	به دره‌های ناهموار سطح ماه نفوذ کنند	bumpy surface
planetary scientists	scientists that study planets	دانشمندانی که سیاره‌ها را مطالعه می‌کنند	space science; aerospace scientist
smashed into Earth	hit Earth very hard while moving fast	به‌سختی با زمین برخورد کرد	collide; crash
Earth spins on its own axis	Earth rotates on a particular invisible line	زمین روی محور خودش می‌چرخد	Earth's axial tilt
it has a natural tendency to wobble	a natural inclination toward shaking	تمایل طبیعی به تکان‌خوردن	natural disposition
the varying pull from other bodies	the different forces from other physical structures, such as planets, that try to move Earth toward themselves	کشش/جاذبه های مختلف از سوی دیگر سیاره ها	gravitational pull opposite of pull: repel
rotational instabilities	the lack of stability in the act of rotating or spinning	نااستواری های مربوط به چرخش (زمین)	opposite: stability
Earth's climatic zones	divisions of the Earth's climates into general climate zones according to average temperatures and average rainfall	مناطق اقلیمی زمین	climate (noun)

other words/ structures	meaning	translation	other structures
physique	the shape and size of a human body	ساختمان بدن	body; figure
without a shadow of a doubt	with certainty; for sure	بدون ذره‌ای شک	beyond a/any doubt
I beg to differ	I politely disagree	با شما مخالفم (مؤدبانه)	similar: agree to disagree
it is on my bucket list	it is on my wish list; a bucket list is some experiences or achievements that a person hopes to have or accomplish during their lifetime	در لیست آرزوهایم است	to-do list
a striking coincidence	an interesting occurrence of two or more events at the same time by chance	یک اتفاق جالب	happy accident
vibration	continuous quick shaking movement	لرزش	quivering; trembling



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